

Paper outline (JCP)

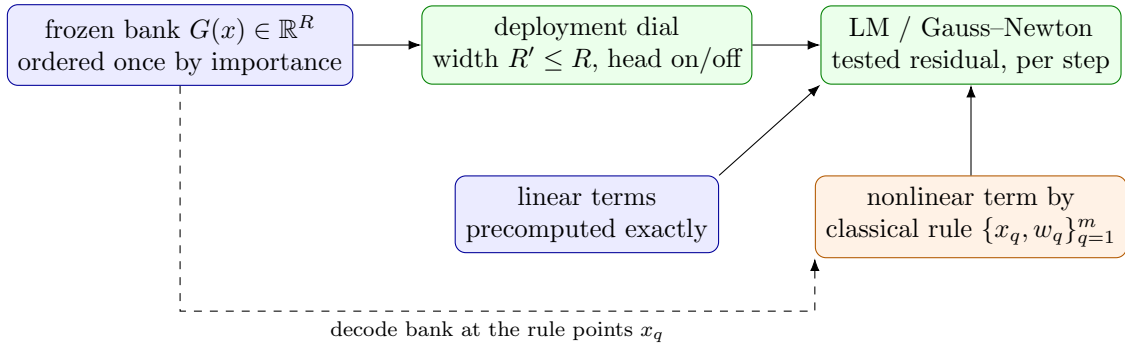
Continuous coordinate-network bases for Petrov–Galerkin reduced models:
deployment-time tunability and classical-quadrature hyper-reduction

Working outline, 2026-10-06 — planning document, no new results

How to read this. Every item is tagged with its evidence status: [have] exists at full scale and can be reused once re-scored; [re-run] exists but must be redone under the new protocol (second-order references, sealed cohorts); [new] not yet run; [theory] an analysis to write. Experiment ids E0–E9 refer to §13. No numbers are given: all paper numbers will be generated from run JSONs. Sources: `reports/2026-10-06-jcp-offmesh-paper-plan.md` and its appendices, the discarded ICLR manuscript `paper.latex/main.tex`, and Hari’s off-mesh quadrature study.

One-sentence thesis

A reduced trial basis that is a *continuous, importance-ordered* function of position, $u(x) = G(x)c$, gives two properties no mesh-bound basis has. **(i) Ordered:** a single trained model exposes an accuracy/cost dial at deployment, by choosing the bank width R' and whether the nonlinear head is used. **(ii) Continuous:** the nonlinear term is an integral, so it can be hyper-reduced by *classical* quadrature rather than by quadrature fitted to snapshots on the mesh. This removes per-mesh fitting and the mesh stencil’s consistency error.



Title options

1. Continuous coordinate-network bases for Petrov–Galerkin reduced models: deployment-time tunability and classical-quadrature hyper-reduction (*recommended*)
2. Tunable neural-field reduced-order models with classical quadrature in place of empirical hyper-reduction
3. One model, many operating points: ordered neural-field bases and off-mesh quadrature for reduced-order models

Highlights (JCP, 3–5 bullets, each ≤ 85 characters)

- One trained neural-field basis gives an accuracy/cost dial at deployment.
- Classical Gauss and lattice rules replace snapshot-fitted hyper-reduction.
- Off-mesh evaluation removes the mesh stencil’s error from the reduced model.
- Rule convergence is explained by integrand regularity; sparse grids fail.
- Fair tests against POD+ECSW/DEIM/GNAT and efficient full-order solvers.

Abstract skeleton (≤ 250 words)

Problem: reduced models are trained for one operating point, and hyper-reduce nonlinear terms by fitting points to snapshots on a mesh. Method: a coordinate-network bank with exact Dirichlet enforcement, ordered by importance; a least-squares Petrov–Galerkin residual with fixed sine tests and exactly precomputed linear terms; the nonlinear term integrated by tensor Gauss or rank-1 lattice rules at points that ignore the mesh. Findings: [dial range on Burgers 2D/3D]; [continuum-target error independent of the mesh, against a mesh-tensor incumbent]; [storage reduction]; [rule ranking explained by theory]; [comparison with POD+ECSW at matched cost]. Limits, stated plainly: [cost relative to tuned full-order solvers by mesh]; box domains; head setting status; linear PDEs where POD is the stronger choice.

1 Introduction

- Reduced models in many-query settings; the two costs addressed here: one fixed operating point per trained model, and hyper-reduction fitted per mesh.
- Neural-manifold ROMs (Lee–Carlberg; Kim et al.) remain mesh-bound. Continuous decoders (CROM, SNF-ROM, Weder et al.) evaluate off the mesh but use unweighted or uniform points, with no rule theory and no tunability. *Credit every “first” correctly.*
- Contributions:
 - (C1) an importance-ordered coordinate-network bank whose width and head are a deployment-time dial [re-run];
 - (C2) classical quadrature (Gauss, rank-1 lattices) replacing fitted hyper-reduction for the tested nonlinear term [have][re-run];
 - (C3) analysis: rule error from integrand regularity, the mesh-target vs continuum-target decomposition, a Strang-type ROM error bound, and a cost model [theory];
 - (C4) full-scale evidence on Burgers 2D/3D, a non-polynomial reaction–diffusion problem, NS 3D and linear problems, against standard hyper-reduced ROMs and efficient full-order solvers [new].
- Limitations previewed here, not hidden in the conclusion.

2 Related work

- LSPG and nonlinear-manifold ROMs (Carlberg et al.; Lee–Carlberg; Kim–Choi–Widemann–Zohdi; Romor et al.; Diaz et al.).
- Hyper-reduction: EIM/DEIM/Q-DEIM, GNAT, ECSW, EQP (Yano–Patera), ECM/CECM; “a priori hyperreduction” (Ryckelynck) — avoid that phrase.
- Continuous and neural-field ROMs: CROM, LiCROM, neural stress fields, Simplicits, SNF-ROM, Weder–Schwerdtner–Peherstorfer, CoLoRA, CNF-ROM.
- Adaptivity and tunability in ROMs (error-driven basis enrichment; adaptive h-refinement of ROMs); neural operators (one paragraph only).
- Quadrature theory: Gauss for analytic integrands (Trefethen), lattice rules (Sloan–Joe; Dick–Kuo–Sloan), sparse grids (Gerstner–Griebel; Bungartz–Griebel); VPINN quadrature analysis (Berrone–Canuto–Pintore).
- Open check before submission: arXiv 2505.14595 and 2507.07830 (not yet read).

3 The reduced model

- **Bank.** Coordinate network with random Fourier features and SiLU; exact Dirichlet factor; trained on snapshots; ordering by importance (column rotation/compression) giving a nested basis in which the first R' columns carry the most snapshot energy [have].
- **Trial manifold.** Linear span $u = G_{R'}(x)c$; optional nonlinear head $c = h(z)$, $z \in \mathbb{R}^k$; the dial is (R', head) .
- **Petrov–Galerkin residual.** $M \approx 4R'$ tensor sine tests ψ_{ab} ; backward Euler; $r(c) = D(Ac - p + \Delta t(N(c) + \nu \Lambda Ac))$, with $A = \Phi^\top G$ and Λ precomputed exactly; Levenberg–Marquardt, with first-order optimality as the stopping test.
- **Matrix-free implementation** in JAX; forward-mode head Jacobian.
- Notation follows the earlier manuscript (K , F , $R(u)$, J_D , N_{eq} , span $c \in \mathbb{R}^{R'}$).

4 Hyper-reduction of the nonlinear term

- The tested term $N_{ab}(c) = \int_{\Omega} \psi_{ab} u (u_x + u_y) dx$, generalised to $\int \psi F(u, \nabla u)$.
- **Mesh-target rules:** dense stencil, mesh sub-lattice, NNLS empirical quadrature (ECSW-type), exact quadratic tensor (MR^2 storage).
- **Continuum-target rules:** tensor Gauss–Legendre p^d , rank-1 Fibonacci/CBC lattices with a random shift; Sobol and Smolyak as controls only.
- Off-mesh assembly $N(c) = P^\top [(Bc) \odot (Dc)]$, with $B = G(x_q)$ and $D = \partial G(x_q)$ decoded once offline; storage $m(2R' + M)$.
- Data-free error indicator: rule doubling and spread over random shifts. Called an *indicator*, not a certificate, unless E7 supports more.
- All-continuum variant: linear terms, initial projection and outputs also by quadrature, so the online path never touches the mesh [new].

5 Analysis [\[theory\]](#)

1. **Gauss rules.** Geometric convergence for an analytic decoder via Bernstein ellipses. The explicit strip-width bound must be re-derived (the audit found the draft’s layer-wise argument wrong).
2. **Lattice rules.** The integrand vanishes to at least second order at the walls; this gives a *conditional algebraic upper bound*. Observed slopes are reported separately. The tent transform makes this integrand smoother (correcting the “adds a kink” explanation).
3. **Sparse grids.** Exactness on the full tensor space needs at least n^d nodes; diagonal sine modes fall in the discarded part. The error lower bound is stated as heuristic.
4. **Two targets.** The sign-upwind stencil equals the continuum term minus an $O(h)$ numerical-diffusion term, so mesh rules and continuum rules converge to different limits. Rollout mesh invariance needs, in addition, stability and the same solution branch.
5. **ROM error bound.** One-step perturbation via Newton–Kantorovich; accumulation over steps (quadrature error enters with Δt); Strang-type total: basis floor + projection + time + linear terms + quadrature (+ $O(h)$ for mesh rules). Stability constant measured, not proved.
6. **Tunability.** Monotonicity of the best-fit error in R' for an ordered bank; cost of a query as a function of (R', m, M) .
7. **Cost model.** Flops and bytes per residual and Jacobian for each rule; the crossover point count against the tensor; the parts that remain $O(N)$.

6 Numerical study I: the dial (tunability)

- Burgers 2D: error and cost along the (R', head) ladder at several meshes, against order-verified references [\[re-run\]](#).
- Burgers 3D and NS 3D ladders [\[re-run\]](#).
- Poisson and heat (2D/3D): the dial on linear problems, where the top rung is the linear span; presented as validation, with POD at matched width shown alongside [\[re-run\]](#).
- **Fig. 1:** error–cost curves of the dial against the full-order solver’s own cost–accuracy curve. **Tab. 1:** operating points per PDE.

7 Numerical study II: quadrature

- Rule ladders (ρ vs m), 2D and 3D, every rule family, controls must fail against the converged target [\[have\]](#)[\[re-run\]](#).
- Measured slopes against the analysis; decoder ablation (RFF, SIREN, RBF, fixed sine bank, POD+cubic, POD+bilinear) [\[new\]](#).
- Shift and scramble distributions; error-indicator effectivity and coverage [\[new\]](#).
- **Fig. 2:** ρ vs m (2D, 3D). **Fig. 3:** decoder ablation. **Tab. 2:** rule sizes needed per mesh and width.

8 Numerical study III: mechanism

- Gap between mesh and continuum targets: slope 1 for the first-order stencil, slope 2 for a second-order stencil [\[new\]](#).
- Controls separating the causes: mesh nodes with the analytic gradient; a bank trained on coarsest-mesh data only; the all-continuum arm [\[new\]](#).
- Allen–Cahn as the no-gap control [\[new\]](#).
- **Fig. 4:** gap vs h . **Fig. 5:** ROM error vs mesh for each arm.

9 Numerical study IV: reduced solves and baselines

- Burgers 2D (256^2 – 4096^2) and 3D (64^3 – 256^3) on sealed test cohorts against second-order references; per-case error budget with each term measured separately [\[re-run\]](#).
- Non-polynomial reaction–diffusion (e^u) and Allen–Cahn at full scale [\[new\]](#).
- Baselines at matched online dimension: POD-LSPG/Galerkin with ECSW, Q-DEIM and GNAT (native solvers); POD with exact tensor; best-practice EQ on our bank; CROM-style sampler; earlier NM-ROM [\[new\]](#).
- The 2D head setting: diagnosis, or documented limitation [\[new\]](#).
- NS 3D: off-mesh against the pseudo-spectral full-order model (cost only; no stencil gap) [\[new\]](#).
- **Tab. 3:** accuracy and cost, all methods, all PDEs. **Fig. 6:** Pareto plot per PDE.

10 Cost

- Per-Jacobian time, flops, bytes and memory; roofline argument [\[have\]](#)[\[re-run\]](#).
- Solve time and full query time against mesh size [\[have\]](#)[\[re-run\]](#).
- Speed-up against validated settings of named efficient solvers (Newton–BiCGStab, Newton–GMRES, IMEX with CG, a second-order scheme), coarse-mesh full-order runs included; always printed with the accuracy pair [\[new\]](#).
- Offline cost (data, training, rule construction, NNLS fits) and break-even query counts [\[new\]](#).
- **Fig. 7:** time vs mesh size. **Tab. 4:** speed-ups with accuracy pairs.

11 Discussion and limitations

- Where the method wins (large meshes, many queries, nonlinear terms without a tractable tensor) and where it does not (coarse meshes; linear problems, where POD is the better choice).
- Box domains with homogeneous Dirichlet conditions; the route to complex geometry (mapped or composite rules, distance-function factors).
- The head setting; smooth solutions only; projection and output decoding remain $O(N)$ unless the all-continuum path is used.

12 Conclusion

Data and code availability

Zenodo snapshot of the code mirror; run JSONs; frozen selection hashes; scripts that generate every table; bank weights or seeds to regenerate them.

Appendices

- A. Proofs.
- B. Bank architecture, training and ordering.
- C. Reference solutions: second-order schemes, observed order, Richardson uncertainty.
- D. Cohorts, sealing and selection protocol.
- E. Timing protocol (paired A–B–A runs, drift gates, injected-fault control).
- F. Full rule ladders and per-mesh tables.
- G. Baseline implementations and tuning budgets.
- H. Reproducibility checklist.

13 Experiment map

id	experiment	feeds
E0	Order-verified second-order references; new sealed and out-of-distribution cohorts	all
E1	Rule ladders and slopes; polynomial-bank machinery check	§6, §5
E2	Mechanism controls: second-order stencil, mesh-node analytic gradient, coarse-trained bank, all-continuum	§7
E3	End-to-end Burgers 2D/3D on sealed cohorts; error budget; head diagnosis	§8
E4	Cost: roofline, flat-in- N , validated full-order comparators, offline cost	§9
E5	Baselines: POD+ECSW/DEIM/GNAT, POD+tensor, EQ, CROM-style	§8
E6	Generality: e^u reaction–diffusion, Allen–Cahn, decoder ablation	§6, §8
E7	Robustness: shifts, error indicator, Δt and R' sweeps	§6
E8	Tunability ladders re-run (Burgers 2D/3D, NS 3D, Poisson, heat) on the new references	§5
E9	NS 3D off-mesh and cost	§8, §9

14 Claims we will not make

- “First off-mesh” or “first mesh-independent” ROM; “a priori hyperreduction”.
- A speed-up headline, or any speed-up without its accuracy pair.
- “More accurate than the full-order model” without naming the scheme, the mesh and the reference.
- “Certified” error, unless E7 supports it.
- “Nonlinear manifold” as the main claim, unless the head setting is confirmed.
- Any number carried over from the ICLR draft without re-running it under the new protocol.